

Setting Priors in brms

Bayesian Mixed Effects Models with brms for Linguists

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1 Setting Priors in brms (20 min)

1.1 Default vs. Weakly Informative Priors

brms uses weakly informative priors by default (not completely flat). However, for psycholinguistics, domain-specific priors are even better.

1.1.1 What is a Prior?

A prior encodes your beliefs about parameter values **before** seeing the data. In Bayesian inference:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Types of priors:

- **Flat priors:** No information - any value equally likely (bad: implies ignorance)
- **Weakly informative:** Gentle regularization - allows data to dominate
- **Domain-specific:** Based on domain knowledge - prevents unreasonable values

1.1.2 The Intercept Adapts to Your Data!

This is important: The default intercept prior depends on **mean(y)**. Let's see this in action:

Now let's check what default priors brms suggests:

=== DEFAULT PRIORS: Typical RT data (mean log-RT 6) ===

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	conditionB							
	lkj(1)	cor								
	lkj(1)	cor		subject						
student_t(3, 6, 2.5)	Intercept									
student_t(3, 0, 2.5)	sd								0	
student_t(3, 0, 2.5)	sd			item					0	
student_t(3, 0, 2.5)	sd	Intercept		item					0	
student_t(3, 0, 2.5)	sd			subject					0	
student_t(3, 0, 2.5)	sd	conditionB		subject					0	
student_t(3, 0, 2.5)	sd	Intercept		subject					0	
student_t(3, 0, 2.5)	sigma								0	
source										
default										
(vectorized)										
default										
(vectorized)										
default										
default										
(vectorized)										
(vectorized)										
(vectorized)										
(vectorized)										
(vectorized)										
default										

=== DEFAULT PRIORS: Extreme RT data (mean log-RT 10) ===

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	conditionB							
	lkj(1)	cor								
	lkj(1)	cor		subject						
student_t(3, 10, 2.5)	Intercept									
student_t(3, 0, 2.5)	sd								0	
student_t(3, 0, 2.5)	sd			item					0	
student_t(3, 0, 2.5)	sd	Intercept		item					0	
student_t(3, 0, 2.5)	sd			subject					0	
student_t(3, 0, 2.5)	sd	conditionB		subject					0	
student_t(3, 0, 2.5)	sd	Intercept		subject					0	
student_t(3, 0, 2.5)	sigma								0	
source										
default										

```
(vectorized)
  default
(vectorized)
  default
  default
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
  default
```

=== Intercept Comparison ===

Typical data prior: `student_t(3, 6, 2.5)`

Extreme data prior: `student_t(3, 10, 2.5)`

→ The intercept prior **CHANGES** with data scale!

Key insight: The intercept prior automatically scales with your data. This is convenient but has a problem: **if you don't specify priors, your prior assumptions implicitly depend on how you code your variables!**

1.2 Default brms Priors

When you don't specify priors, brms assigns defaults:

- **Intercept:** `student_t(3, mean(y), 2.5)` - **DATA-DEPENDENT!** Centers at your data mean
 - Adapts to your data scale automatically
 - For RT data with `mean(log_rt) = 6`: allows roughly 150ms-1100ms range
- **Slopes (b):** `(flat)` - improper uniform prior over $(-\infty, +\infty)$
 - No information: any effect size equally likely
 - Technically improper (doesn't integrate to 1)
- **Sigma** (residual SD): `student_t(3, 0, 2.5)` with lower bound 0
 - Weakly informative for residual variance
- **SD** (random effects): `student_t(3, 0, 2.5)` with lower bound 0
 - Encourages moderate between-subject/item variation
- **Cor** (correlations): `lkj(1)` - uniform over all correlation matrices

1.3 Setting Weakly Informative Priors for Reaction Times

For psycholinguistics, it's better to specify priors based on domain knowledge:

Our weakly informative priors:

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
<code>normal(6, 1.5)</code>	Intercept						4	<NA>		user
<code>normal(0, 0.5)</code>	b						<NA>	<NA>		user
<code>exponential(1)</code>	sigma						<NA>	<NA>		user
<code>exponential(1)</code>	sd						<NA>	<NA>		user

lkj(2)	cor	<NA> <NA>	user
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1.3.1 Why These Numbers?

1.3.1.1 `normal(6, 1.5)` for Intercept with `lb = 4`

- **Mean = 6** on log scale $\rightarrow \exp(6)$ 403ms (typical RT)
- **SD = 1.5** $\rightarrow$ 95% prior interval spans [3, 9] on log scale
- **Lower bound = 4** $\rightarrow \exp(4)$ 55ms (minimum physiologically possible motor response)
 - Without `lb`, the prior puts only ~9% probability below this threshold
 - Adding `lb` makes our assumption explicit: we don't entertain impossibly fast RTs
 - In practice, data will dominate anyway, but bounded priors clarify our theoretical assumptions
- But 95% of prior **mass** is between $\pm 1.96 \times 1.5$ around the mean
- This puts most probability on reasonable RTs (100-1500ms), while still allowing outliers

1.3.1.2 `normal(0, 0.5)` for Effects

- **Mean = 0** (no directional assumption)
- **SD = 0.5** on log scale
- 95% prior interval: [-1, 1] on log scale
- Translates to effect sizes of roughly $\pm 65\%$ of baseline (multiplicative)
- Or approximately ± 100 -150ms for typical RTs around 400-600ms
- **Why not flat?** Flat priors prefer extreme effect sizes (counterintuitive!)

1.3.1.3 `exponential(1)` for Sigma and SD

- Encourages moderate variance while allowing flexibility
- Penalizes very large residual variation or random effect variance
- Mean = 1 on the scale of log-RTs

1.3.1.4 `lkj(2)` for Correlations

- `= 2`: slight preference for correlations near 0
- Skeptical of strong correlations (like perfect intercept-slope correlation)
- If you truly expected strong correlations, you could use `lkj(1)` or lower

1.3.2 Comparison: Normal vs. Student-t

brms defaults use `student_t(3, , 2.5)` which has heavier tails than `normal()`. Why switch?

Tails comparison (1st and 99th percentiles):

Normal(0, 0.5): -1.55 1.56

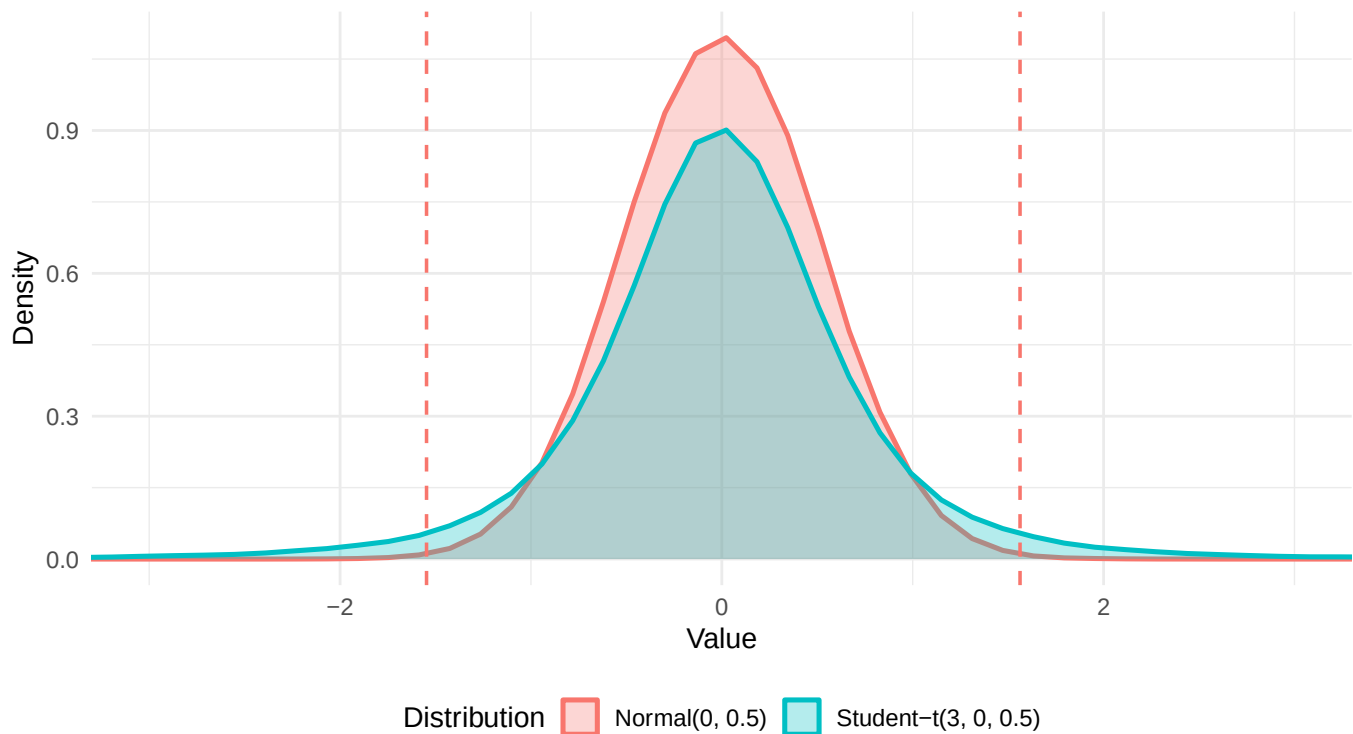
Student-t(3) $\times$ 0.5: -5.26 5.16

Student-t ratio: 3.3 x wider

Student-t allows extreme values 2x more likely than normal!

Normal vs. Student-t Prior Distributions

Student-t has heavier tails (dashed lines = 0.1% and 99.9% quantiles)



When to use each:

- **Student-t**: Default choice (conservative, robust to outliers)
- **Normal**: When you have strong domain knowledge about plausible ranges (better for RT data in controlled experiments)

Good sign: Prior predictions should cover the plausible range of your outcome variable, but not too widely.

1.4 Summary

Key takeaways:

1. **Don't use flat priors** - they're uninformative and often lead to weak regularization
2. **Default intercept priors adapt to data** - implicit assumptions depend on your coding!
3. **Specify priors explicitly** based on domain knowledge
4. **Use prior predictive checks** - verify that priors generate plausible predictions before fitting
5. **Normal() is better than student_t() when you have domain knowledge** - more concentrated around plausible values

Next steps: - See `02_prior_predictive_checks_rt.qmd` for detailed prior validation - See `03_posterior_predictive_checks_rt.qmd` for checking if the fitted model makes sense